TOPIC 11

Scaling: Normalisation and Standardisation: Explained

The original class slides, with a plain explanation under each one

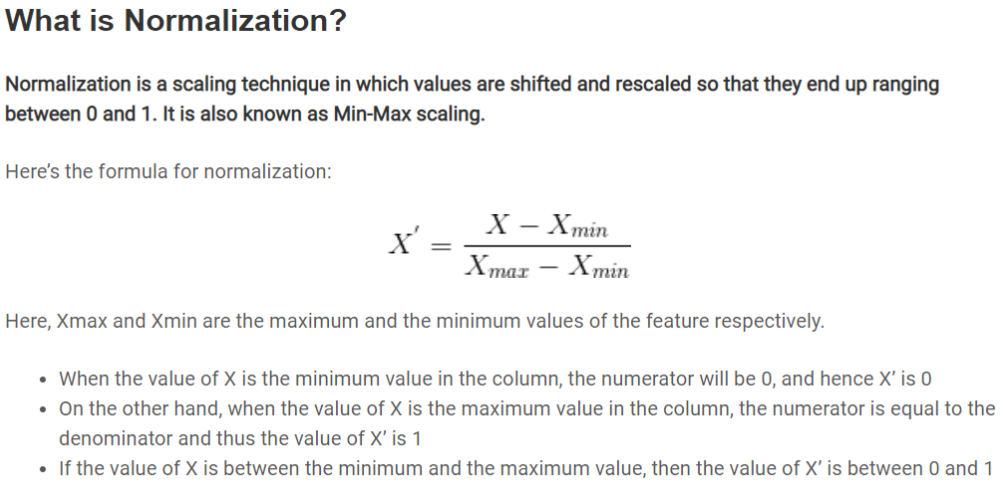
> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

Features are often measured in very different units (age in tens, salary in tens of thousands). Many algorithms are thrown off by that. **Scaling** puts the features on a comparable scale. These 3 slides explain the two main methods, **normalisation** and **standardisation**, and how to choose between them.

Words you will meet

Word	Plain meaning
Normalisation (min-max scaling)	Rescales values to lie between 0 and 1
Standardisation (z-score)	Rescales values to have mean 0 and standard deviation 1
X_min / X_max	Smallest / largest value of the feature
μ (mu)	The mean (average) of the feature
σ (sigma)	The standard deviation of the feature: how spread out the values are
Gaussian distribution	The bell-shaped distribution
Data leakage	Test-set information accidentally used during training, which makes results look better than they are

Slide 1: Normalisation (min-max scaling)



What is Normalization?

Normalization is a scaling technique in which values are shifted and rescaled so that they end up ranging between 0 and 1. It is also known as Min-Max scaling.

Here's the formula for normalization:

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Here, Xmax and Xmin are the maximum and the minimum values of the feature respectively.

- When the value of X is the minimum value in the column, the numerator will be 0, and hence X' is 0
- On the other hand, when the value of X is the maximum value in the column, the numerator is equal to the denominator and thus the value of X' is 1
- If the value of X is between the minimum and the maximum value, then the value of X' is between 0 and 1

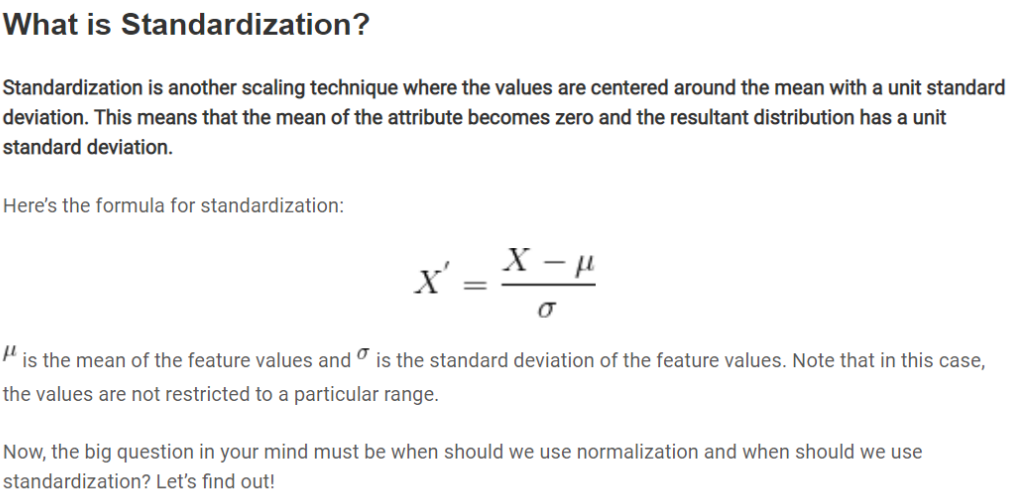
Slide 1

Normalisation shifts and rescales values so they end up between **0 and 1**. The formula subtracts the smallest value and divides by the range (largest minus smallest). The three bullets explain what happens at each end:

- The **smallest** value gives a numerator of 0, so X′ = **0**
- The **largest** value makes the numerator equal to the denominator, so X′ = **1**
- Any value **between** them gives X′ between **0 and 1**

$$X' = \frac{X - X_{\text{min}}}{X_{\text{max}} - X_{\text{min}}}$$

Slide 2: Standardisation (z-score)



What is Standardization?

Standardization is another scaling technique where the values are centered around the mean with a unit standard deviation. This means that the mean of the attribute becomes zero and the resultant distribution has a unit standard deviation.

Here's the formula for standardization:

$$X' = \frac{X - \mu}{\sigma}$$

μ is the mean of the feature values and σ is the standard deviation of the feature values. Note that in this case, the values are not restricted to a particular range.

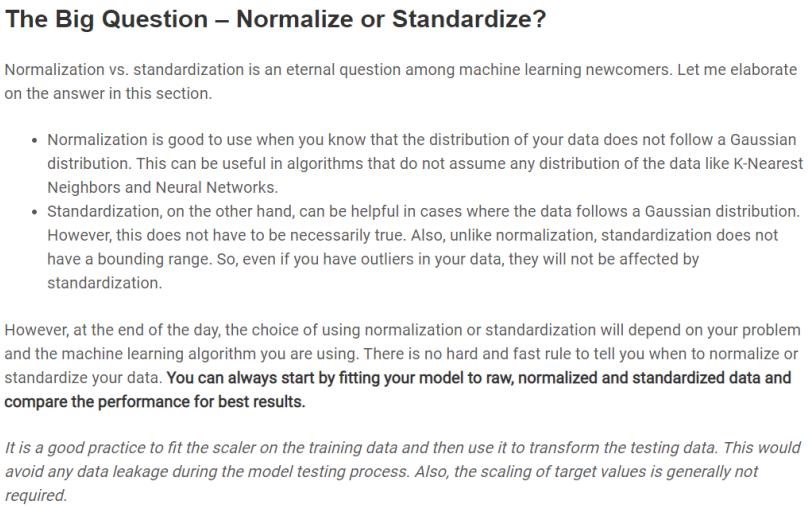
Now, the big question in your mind must be when should we use normalization and when should we use standardization? Let's find out!

Slide 2

Standardisation centres the values around the **mean** with a **unit standard deviation**: after the change the mean of the feature is **0** and the spread (standard deviation) is **1**. Subtract the mean μ from each value and divide by the standard deviation σ. The slide stresses that, unlike normalisation, **the values are not restricted to a particular range**: a standardised value can be −3, 0.5 or 4.

$$X' = \frac{X - \mu}{\sigma}$$

Slide 3: Which one should I use?



The Big Question – Normalize or Standardize?

Normalization vs. standardization is an eternal question among machine learning newcomers. Let me elaborate on the answer in this section.

- Normalization is good to use when you know that the distribution of your data does not follow a Gaussian distribution. This can be useful in algorithms that do not assume any distribution of the data like K-Nearest Neighbors and Neural Networks.
- Standardization, on the other hand, can be helpful in cases where the data follows a Gaussian distribution. However, this does not have to be necessarily true. Also, unlike normalization, standardization does not have a bounding range. So, even if you have outliers in your data, they will not be affected by standardization.

However, at the end of the day, the choice of using normalization or standardization will depend on your problem and the machine learning algorithm you are using. There is no hard and fast rule to tell you when to normalize or standardize your data. **You can always start by fitting your model to raw, normalized and standardized data and compare the performance for best results.**

It is a good practice to fit the scaler on the training data and then use it to transform the testing data. This would avoid any data leakage during the model testing process. Also, the scaling of target values is generally not required.

Slide 3

The slide gives a rule of thumb, and then an honest caveat:

- **Normalisation** is good when you know the data does **not** follow a bell-shaped (Gaussian) distribution. It is useful for algorithms that make no assumption about the distribution, such as **K-Nearest Neighbours** and **neural networks**.
- **Standardisation** can be helpful when the data **does** follow a bell shape (though it does not have to). It has **no bounding range**, so, as the slide says, extreme values are not squeezed together the way min-max scaling squeezes them (they still influence the mean and standard deviation, so they are not completely ignored).
- **There is no hard rule.** You can always fit your model on raw, normalised and standardised data and compare the results.
- **Good practice:** fit the scaler on the **training data only**, then use it to transform the test data, so no test information leaks into training. Scaling the **target** is generally not required.